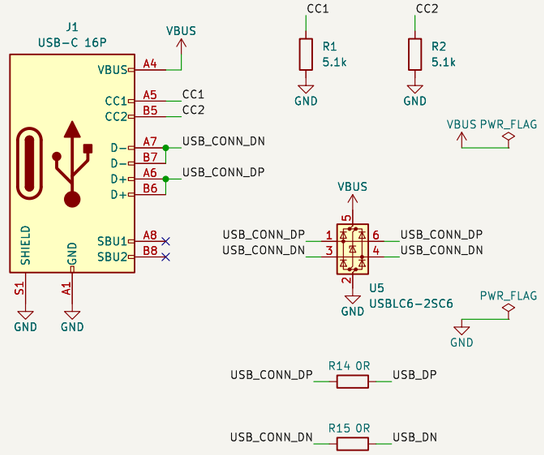
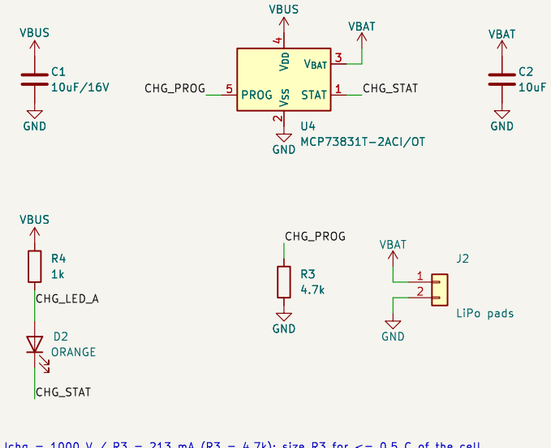
USB-C + ESD



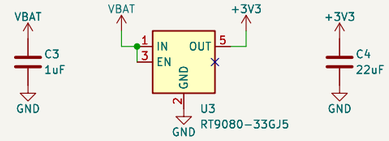
USB 2.0 FS goes to the ESP32-H2 USB-Serial/JTAG pins (GPIO26 = D-, GPIO27 = D+).
R1, R2: Rd 5.1k = USB-C sink. R14, R15: 0R placeholders for EMI tuning.

LiPo charger



Ichg = 1000 V / R3 = 213 mA (R3 = 4.7k): size R3 for <= 0.5 C of the cell.
D2 lit = charging (STAT low). No power-path: the load stays on the battery.
J2 = solder pads on the bottom side: pad 1 = BAT+, pad 2 = BAT-. CHECK THE POLARITY.

3V3 LDO

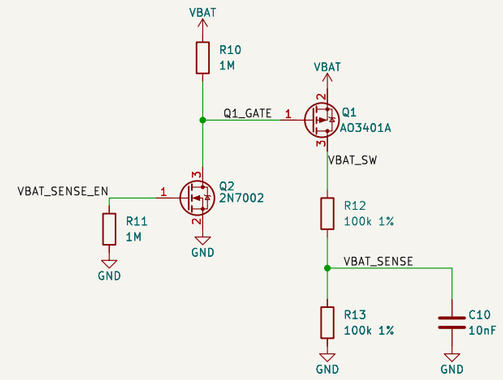


RT9080-33GJ5: Iq 2 uA, 600 mA. EN tied to VIN (always on).
Pinout 1=VIN 2=GND 3=EN 4=NC 5=VOUT, same as
TPS7A0533PDBV and XC6220B331MR-G (alternates).

Design notes

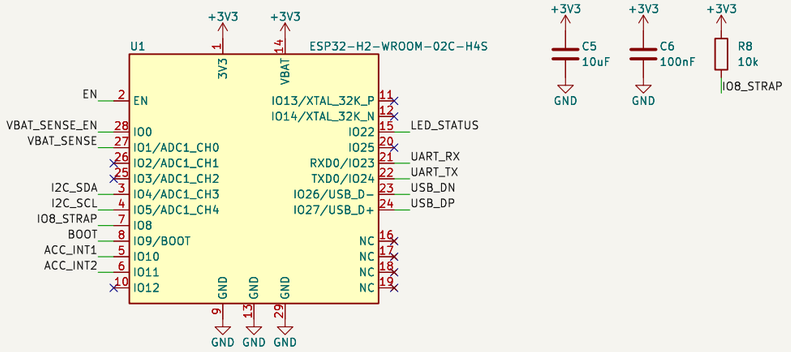
PCB: 4 layers recommended (sig / GND / PWR / sig), 1.6 mm.
U1: antenna at the board edge, no copper on any
layer under or beside it (keep-out is in the footprint).
Solid GND plane under everything else; stitch vias.
USB D+/D-: route as a pair, short, over GND.
U2: away from the board edge, buttons and battery
connector; no vias or traces under the LGA.
Battery leads and C2 close to J2 / U4.
This sheet is generated: edit hardware/gen/gen_kicad.py.

Battery sense (switched divider)



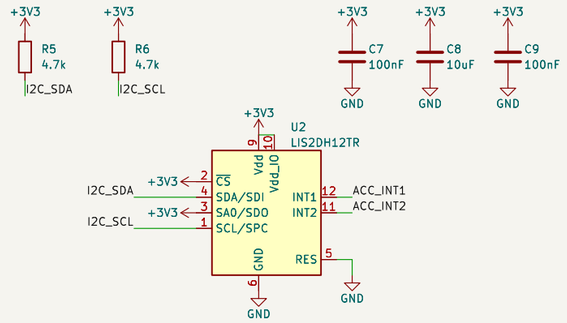
VBAT_SENSE = VBAT / 2 while VBAT_SENSE_EN (GPIO0) is high; settle >= 3 ms.
GPIO0 low or floating (deep sleep): R11 keeps Q2 off, R10 keeps Q1 off,
so the divider draws only leakage. ADC input: GPIO1 = ADC1_CH0.

MCU ESP32-H2-WROOM-02C



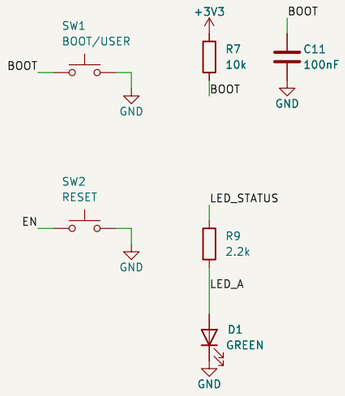
Strapping: GPIO8 pulled up (R8); GPIO9 = BOOT, pulled up (R7), SW1 pulls it low.
GPIO25 (JTAG source strap) is unused: it is ignored with the default eFuses.
EN: the 10k / 1uF power-up RC is inside the module. VBAT pin = 3V3 (module default).
GPIO10 / GPIO11 / GPIO9 are LP IOs (GPIO7..14) and can wake the chip from deep sleep.
Decoupling C5 + C6 right at module pin 1. Keep the antenna area free of copper.

Accelerometer LIS2DH12



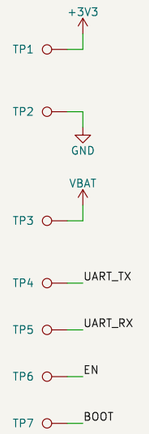
I2C address 0x19 (SA0 high). CS high selects I2C. RES (pin 5) to GND.
INT1 -> GPIO10: FIFO watermark, wakes the MCU. INT2 -> GPIO11: reserved.
The sensor stays powered in sleep (about 2 uA at 1 Hz). C7 / C9 right at the pins.

Buttons / LED



SW1: BOOT / user button (GPIO9, active low).
SW2: reset (EN to GND).
D1: status LED on GPIO22, active high.

Test points



1.0 mm pads for pogo pins:
UART0 console, EN + BOOT
for flashing, supplies.

Wearable worker-activity badge: Zigbee sleepy end device
Generated by hardware/gen/gen_kicad.py - edit the generator, not this file
haxeko/h2-zigbee-lis2dh12

Sheet: /
File: activity-badge.kicad_sch

Title: Activity Badge - ESP32-H2 + LIS2DH12

Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1